19. The terminal potential difference of a cell is 19 V when a current of 1·0 A flows in the circuit. It reduces to 17 V when the current supplied by the cell is 3·0 A. Find the emf and internal resistance of the cell. *2*

22. A cell of emf E and internal resistance r is connected to a variable resistance R. Draw plots showing the variation of (a) terminal voltage V with R, and (b) V with current I, in the circuit. *2*

20. Write two differences between the emf and terminal potential difference of a cell. What is the most important precaution that one should take while drawing current from a cell ? *2*

1. A battery of emf E and internal resistance r is connected to an external circuit. The potential drop within the battery is proportional to :

(a) current in the circuit

(b) total resistance of the circuit

(c) emf of the battery

(d) power dissipated in the circuit

17. *Assertion (A) :* The internal resistance of a cell is constant.

Reason (R) : Ionic concentration of the electrolyte remains same during use of a cell.

13. The emf and internal resistance of a cell are E and r respectively. It is connected across an external resistance R = 2r. The potential drop across the terminals of the cell will be :

(a) $\frac{E}{4}$

(b) $\frac{E}{2}$

(c) $\frac{2}{3}E$

(d) $\frac{E}{3}$

13. A cell of emf E is connected across an external resistance R. When current 'I' is drawn from the cell, the potential difference across the electrodes of the cell drops to V. The internal resistance 'r' of the cell is **1**

(a) $\left(\frac{E-V}{E}\right)R$

(b) $\left(\frac{E-V}{R}\right)$

(c) $\frac{(E-V)\,R}{I}$

(d) $\left(\frac{E-V}{V}\right)R$

OR

(b) (i) A cell emf of (E) and internal resistance (r) is connected across a variable load resistance (R). Draw plots showing the variation of terminal voltage V with (i) R and (ii) the current (I) in the load. **2**

(ii) Three cells, each of emf E but internal resistances 2r, 3r and 6r are connected in parallel across a resistor R.

Obtain expressions for (i) current flowing in the circuit, and (ii) the terminal potential difference across the equivalent cell. **3**

1. A battery supplies 0.9 A current through a 2 Ω resistor and 0.3 A current through a 7 Ω resistor when connected one by one. The internal resistance of the battery is :

(A) 2 Ω

(B) 1.2 Ω

(C) 1 Ω

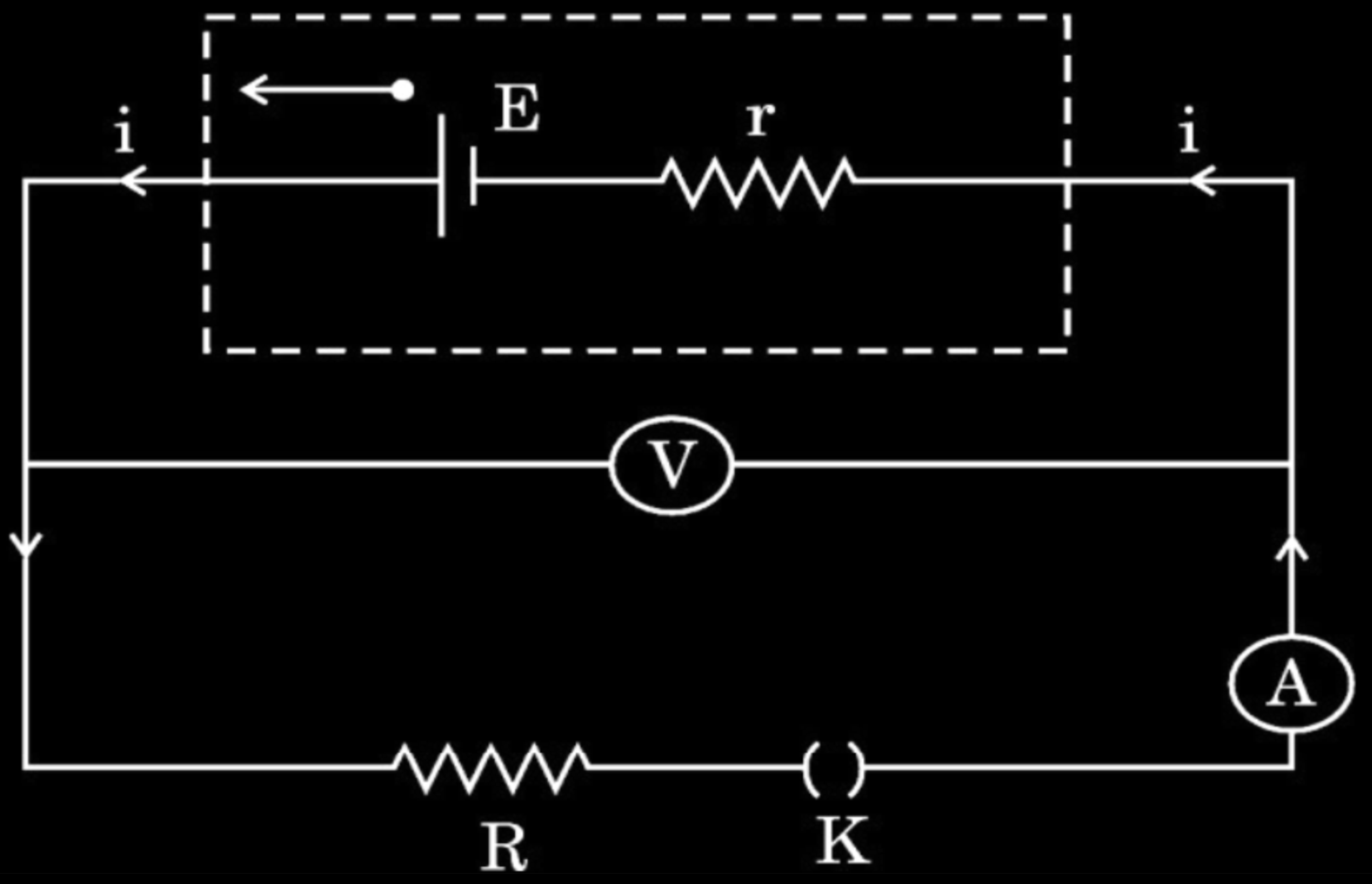
(D) 0.5 Ω

22. A battery of unknown emf E and internal resistance r is connected in a circuit as shown in the figure. When the key (K) is open, the voltmeter reads 10·0 V and ammeter reads 0 A. In the closed circuit, the voltmeter reads 6·0 V and ammeter reads 2·0 A. Calculate : *3*

(a) emf of the battery,

(b) internal resistance of the battery (r), and

(c) external resistance (R).



30. When the terminals of a cell are connected to a conductor of resistance R, an electric current flows through the circuit. The electrolyte of the cell also offers some resistance in the path of the current, like the conductor. This resistance offered by the electrolyte is called internal resistance of the cell (r). It depends upon the nature of the electrolyte, the area of the electrodes immersed in the electrolyte and the temperature. Due to internal resistance, a part of the energy supplied by the cell is wasted in the form of heat.

When no current is drawn from the cell, the potential difference between the two electrodes in known as emf of the cell (ε). With a current drawn from the cell, the potential difference between the two electrodes is termed as terminal potential difference (V).

(i) Choose the ***incorrect*** statement : *1*

(A) The potential difference (V) between the two terminals of a cell in a closed circuit is always less than its emf (ε), during discharge of the cell.

(B) The internal resistance of a cell decreases with the decrease in temperature of the electrolyte.

(C) When current is drawn from the cell then $V = \varepsilon - Ir$.

(D) The graph between potential difference between the two terminals of the cell (V) and the current (I) through it is a straight line with a negative slope.

(ii) Two cells of emfs 2·0 V and 6·0 V and internal resistances 0·1 Ω and 0·4 Ω respectively, are connected in parallel. The equivalent emf of the combination will be : *1*

(A) 2·0 V (B) 2·8 V

(C) 6·0 V (D) 8·0 V

(iii) Dipped in the solution, the electrode exchanges charges with the electrolyte. The positive electrode develops a potential V_+ ($V_+ > 0$), and the negative electrode develops a potential $-(V_-)$ ($V_- \geq 0$), relative to the electrolyte adjacent to it. When no current is drawn from the cell then : *1*

(A) $\varepsilon = V_+ + V_- > 0$ (B) $\varepsilon = V_+ - V_- > 0$

(C) $\varepsilon = V_+ + V_- < 0$ (D) $\varepsilon = V_+ + V_- = 0$

(iv) (a) Five identical cells, each of emf 2 V and internal resistance 0·1 Ω are connected in parallel. This combination in turn is connected to an external resistor of 9·98 Ω. The current flowing through the resistor is : *1*

(A) 0·05 A (B) 0·1 A

(C) 0·15 A (D) 0·2 A

OR

(b) Potential difference across a cell in the open circuit is 6 V. It becomes 4 V when a current of 2 A is drawn from it. The internal resistance of the cell is : *1*

(A) 1·0 Ω (B) 1·5 Ω

(C) 2·0 Ω (D) 2·5 Ω

2. The emf and internal resistance of a cell are 3 V and 0·2 Ω respectively. It is connected to an external resistor of 5·8 Ω. The potential difference across the cell will be :

(A) 1·2 V (B) 2·0 V

(C) 2·9 V (D) 3·2 V

19. (a) A cell is connected across an external resistance 12 Ω and supplies 0.25 A current. When the external resistance is increased by 4 Ω, the current reduces to 0.2 A. Calculate (i) the emf, and (ii) the internal resistance, of the cell. *2*

17. A cell of emf E and internal resistance r is connected across a resistor of variable resistance R. Show graphically the variation of

(a) the terminal voltage across the cell,

(b) the current supplied by the cell,

with R as it is increased from 0 to the maximum value. *2*

17. A cell of emf E and internal resistance r is connected to an external variable resistance R. Plot a graph showing the variation of terminal voltage V of the cell as a function of current I, supplied by the cell. Explain how the emf of the cell and its internal resistance can be found from it. *2*

22. (a) Define the terms 'emf' and 'terminal voltage' of a cell. Can the terminal voltage of a cell ever exceed its emf ? Explain. **3**

OR

(b) Define the term current density. Write its SI unit. Derive the equivalent form of Ohm's law $\vec{j} = \sigma \vec{E}$. **3**

OR

(b) (i) A battery of emf E and internal resistance r is connected to a variable external resistance R.

(I) Obtain the expression for current I in the circuit and the value of maximum current the battery can supply.

(II) Obtain the terminal voltage V across the battery and its maximum possible value.

(ii) The above battery sends a current I_1 when $R = R_1$ and a current I_2 when $R = R_2$. Obtain the internal resistance of the battery in terms of I_1, I_2, R_1 and R_2. *3*

2. A battery of e.m.f. 12 V and internal resistance 0.5 Ω is connected to a 9.5 Ω resistor through a key. The ratio of potential difference between the two terminals of the battery, when the key is open to that when the key is closed, is **1**

(A) 1.05 (B) 1

(C) 0.95 (D) 1.1

22. (a) A cell of e.m.f. E and internal resistance r is connected with a variable external resistance R and a voltmeter showing potential drop V across R. Obtain the relationship between V, E, R and r. **3**

(b) Draw the shape of the graph showing the variation of terminal voltage V of the cell as a function of current I drawn from it. How one can determine the e.m.f. of the cell and its internal resistance from this graph ?

17. A battery of emf E and internal resistance r is connected to a rheostat. When a current of 2A is dawn from the battery, the potential difference across the rheostat is $5V$. The potential difference becomes $4V$ when a current of 4A is drawn from the battery. Calculate the value of E and r. **2**